

A
Project Report
On
“Mart Billing System”

Submitted by,

Sr.No	Name of Student	Roll No.
1	Mr. Sangram Babaurao Patil	30
2	Mr. Tanmay Rhushikesh Patil	31
3	Mr. Sushant Suresh Teli	32
4	Mr. Tanmay Tanaji Magdum	33

Under the guidance of

Prof. Dr. D. A. Nikam



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
Dr. J. J. Magdum College of Engineering, Jaysingpur

Academic Year
2023-2024

Dr. J. J. Magdum Trust's

Dr. J. J. Magdum college of Engineering, Jaysingpur.



C E R T I F I C A T E

This is to certify that,

Sr. No.	Name of Student	Roll No.
1	Mr. Sangram Baburao Patil	30
2	Mr. Tanmay Rhushikesh Patil	31
3	Mr. Sushant Suresh Teli	32
4	Mr. Tanmay Tanaji Magdum	33

have satisfactorily completed the Project Phase I entitled “**Mart Billing System**” in fulfillment for award of Bachelor of Engineering Degree in Computer Science & Engineering by Shivaji University, Kolhapur in Academic Year-2023-24.

Project Guide
Prof. D.A. Nikam

Head,
Dept. Of CSE

External Examiner

Principal

**Dr. J. J. Magdum College of Engineering,
Jaysingpur**

Department of Computer Science & Engineering



CERTIFICATE

This is to certify that, the project entitled,

Mart Billing System

is presented before Department Research Committee (DRC) by,

Sr. No.	Name of Student	Roll No.
1	Mr. Sangram Baburao Patil	30
2	Mr. Tanmay Rhushikesh Patil	31
3	Mr. Sushant Suresh Teli	32
4	Mr. Tanmay Tanaji Magdum	33

Under the guidance of Prof. Dr. D.A. Nikam for the academic year 2023-24.

The DRC has consented to give the approval for the said project.

Head,

**Department Research Committee, (DRC)
Department of Computer Science & Engineering**

INDEX

Ch. No.	Topic	Page No.
	Acknowledgement	5
	Abstract	6
	List of Tables	7
	List of Figures	8
Chapter 1	SRS DOCUMENT	9-33
1.1	Introduction	10
1.2	Objectives	11-12
1.3	Outline of dissertation	13-35
Chapter 2	INTRODUCTION	36-38
2.1	Introduction	36
2.2	Objectives	37
2.3	Outline of dissertation	38
Chapter 3	LITERATURE REVIEW	39-43
3.1	Introduction	39
3.2	Brief history of our project	40-43
Chapter 4	METHODOLOGY	44-47
4.1	Introduction	44
Chapter 5	TESTING RESULT	48-52
5.1	Introduction	48
5.2	Table of Test cases	49-52
Chapter 6	CONCLUSIONS	53-54
Chapter 7	REFERENCES	55-57
Chapter 8	LIST OF PUBLICATIONS	58-65

ACKNOWLEDGEMENT

First of all, we would like to thank Dr. Prof. D. A. Nikam who is presently working as a Head of Computer Science & Engineering Department for guiding us through this project work. we are extremely grateful to her for all his invaluable guidance and kind suggestions during our project work. Her ever-encouraging attitude, guidance and whole hearted help, were biggest motivation for us in completing this project work.

We thankful to the founder Chairman Late Dr. J. J. Magdum and the Chairman Mr. Vijayraj J. Magdum of Dr. J. J. Magdum Trust, Jaysingpur, for their encouragement. I am very grateful to Dr. S. S. Admuthe, Campus Director, Dr. Mrs. S. B. Patil, Principal of Dr. J. J. Magdum College of Engineering, Jaysingpur for motivating me for this project work. Also I am thankful to Dr. Prof . D. A. Nikam, Head of Department, Computer Science & Engineering Engineering, for providing necessary facilities for completion of this project work.

Lastly, I thank all the people who have guided and helped us directly or indirectly.

Name Of Student

Signature

Mr. Sangram Baburao Patil

Mr. Tanmay Rhushikesh Patil

Mr. Sushant Suresh Teli

Mr.Tanmay Tanaji Magdum

ABSTRACT

The Smart E-Billing System is an innovative solution designed to streamline and enhance the billing process for businesses across various industries. Traditional paper-based billing systems often result in inefficiencies, delays, and environmental impact. This proposed system leverages modern technologies to automate and digitize the billing process, providing a seamless and eco-friendly solution.

The system offers a range of features designed to streamline the billing process for businesses. It automates the generation of bills, significantly reducing manual effort and minimizing errors typically seen in traditional billing methods. The interface is user-friendly, making it easy for businesses to create, manage, and dispatch bills to customers efficiently. Additionally, the bill templates are customizable, allowing companies to maintain their brand identity and ensure that all communications look professional and consistent. The system also provides real-time updates on billing status, which helps businesses keep track of payments, outstanding balances, and customer interactions effectively. Bills can be delivered through various channels, such as email, SMS, and online portals, adding flexibility and convenience for both businesses and their customers. Furthermore, the system includes secure payment integration, facilitating online payments through reliable payment gateways and decreasing the likelihood of late or missed payments. This comprehensive set of features makes the system an essential tool for modern business operations, ensuring accuracy, efficiency, and security in billing and payment processes.

LIST OF TABLES

Table Number	Table Name
1	Literature review Table
2	Unit Testing
3	Integration Testing
4	System Testing

LIST OF FIGURES

Figure Number	Figure Name	Page No.
1	Proposed System Architecture Diagram	13
2	Flow Chart	15
3	DFD 0 diagram	16
4	DFD 1 diagram	17
5	DFD 2 diagram	18

CHAPTER 1

SRS (Software Requirement Specification)

Software Requirement Specification for Smart Billing System.

Introduction:

This document outlines the software requirements for a smart billing system that utilizes a barcode scanner, a weighing machine, a printer, a web application and a database. The smart billing system is intended to be used in retail stores to improve the efficiency and convenience of shopping by automating the process of tracking and calculating the cost of items

Purpose of this Document:

Shopping is simple but waiting on a bill counter makes shopping too boring and a tedious task. To provide a robust techno-savvy trolley with low-cost, high efficiency and easily adaptable system and to design and develop a system to eliminate time taken at the billing counter in the supermarket.

Scope of this document:

- a) To increase customer satisfaction and reduce human intervention thereby reducing errors faced.
- b) To reduce the average time spent by the customer at the shopping mall.
- c) This system can be also implemented using LI-FI, NFC & other communication systems.
- d) In addition to the product details, nutrition facts of the eatables can be added.
- e) In addition to the product details, nutrition facts of the eatables can be added.
- f) In addition to the product details, nutrition facts of the eatables can be added.

The same system can be used in various places.

Software Requirements

Tools

1. Windows 7 or above
2. Visual studio code (vs code)
3. XAMPP
4. VS Code

Programming Languages

1. Html
2. CSS
3. PHP
4. JavaScript
5. MySQL

HARDWARE REQUIREMENTS

1. Processor: - Intel i5 core
2. RAM: - minimum 4GB
3. Hard Disk : - 50 GB
4. Barcode , Scanner
5. Weighing Machine

SYSTEM ARCHITECTURE

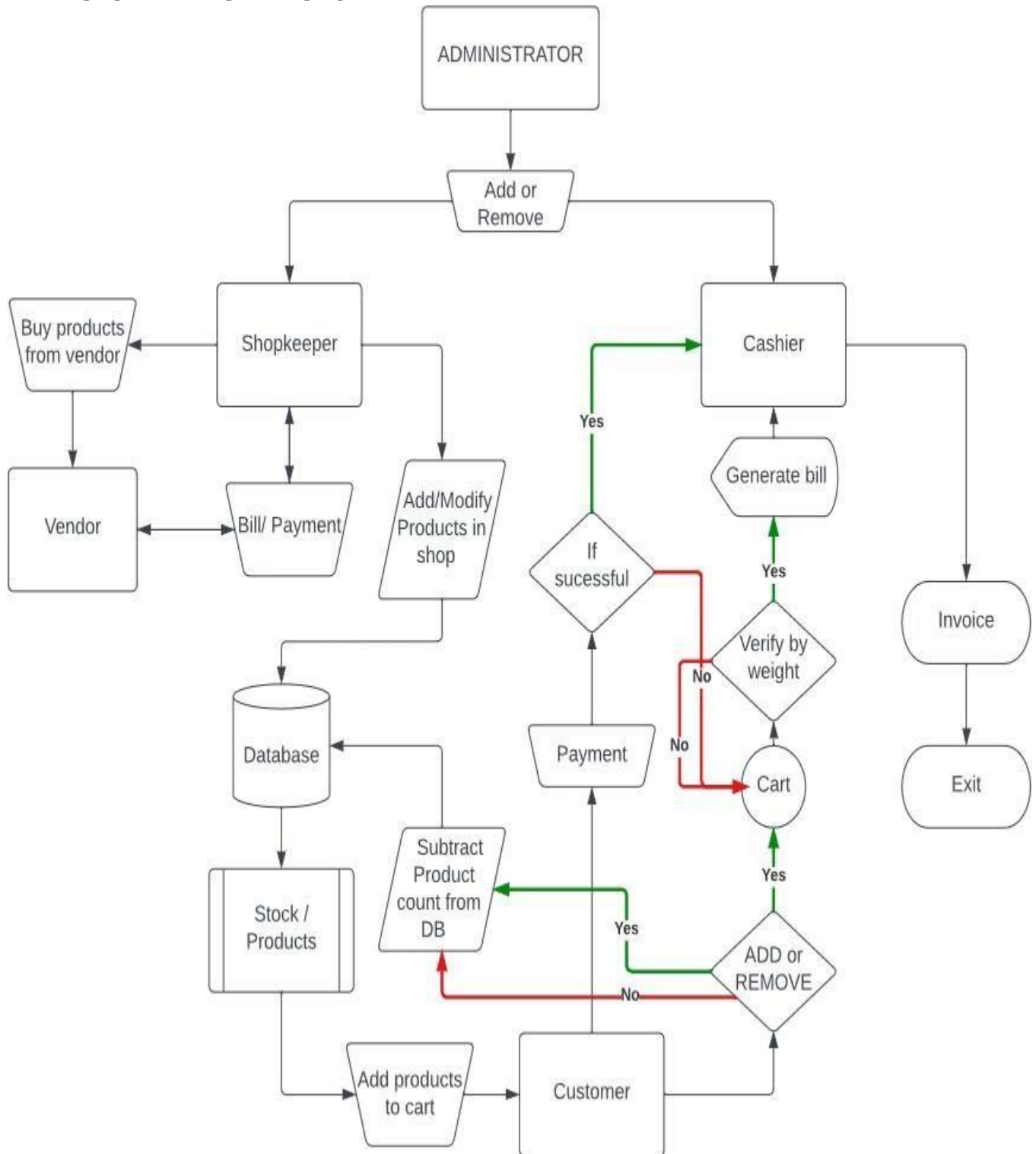


Fig 1-Proposed System Architecture

MODULAR STRUCTURE

In the context of smart billing, a dashboard is a tool that allows users to view and monitor the status of their billing and payment processes. It can display a range of data and metrics related to billing, such as the total sell, the percentage of monthly, percentage of today's purchase, This Month Purchase and Date wise sell.

Purchase Report:

Provides details about purchases made by a user. It includes information such as the date of the purchase, the name and contact information of the vendor or supplier, the items or services purchased, the quantity and cost of each item.

- A. Daily Purchase Report:** Report is a document or report that provides a summary of the purchases made on a specific day.
- B. Product Wise Purchase Report:** Product-wise purchase reports can be useful for a variety of purposes, such as tracking and managing the expenditure of market, analyzing and optimizing purchasing processes, and preparing financial statements.

Vendor:

A vendor is a company or individual that provides goods or services to another specific user.

- A. Add Vendor:** Adding a vendor refers to the process of creating a new record in a system or database for a vendor that a specific user works with. This typically involves entering information such as the vendor's name, contact details, and any relevant information about the products or services they provide.
- B. Vendor List:** A vendor list is a record or list of vendors that an specific user works with. It typically includes information such as the vendor's name, contact details, and any relevant information about the products or services they provide.

Product:

A product is a tangible or intangible item that an specific user sells or provides to its customers.

- A. Add Product:** Adding a product refers to the process of creating a new record in a system or database for a product that a specific user sells or provides. This typically involves entering information such as the product's name, description, price, and any relevant details.
- B. Product List:** A product list is a record or list of products that an specific user sells or provides. It typically includes information such as the product's name, description, price, and any relevant details.

Purchases:

Purchases refer to the process of buying goods or services from a vendor or supplier.

- A) New Purchase:** A new purchase refers to a transaction in which a specific user buys goods or services from a vendor or supplier. This typically involves entering information such as the date of the purchase, the name and contact information of the vendor or supplier, the items or services purchased, the quantity and cost of each item or service, and any applicable taxes or fees.
- B) Purchase List:** The purchase list can be used to keep track of the items being added to the cart, as well as to calculate the total cost of the items. The purchase list can also be used to generate personalized recommendations for the user based on their previous purchases and preferences. In some cases, the purchase list may also be linked to the user's account, allowing them to save and access their list from multiple devices.

Bill:

Admin and Cashier has authority to add a bill and display list

- A) Add Bill:** Adding a bill refers to the process of creating a new record in a system or database for a bill for that a specific user. This typically involves entering information such as the total price, product and any relevant information about the products.
- B) Bill List:** A bill list is a record or list of bills of specific user. It typically includes information such as the total bill, customer name, and any relevant products

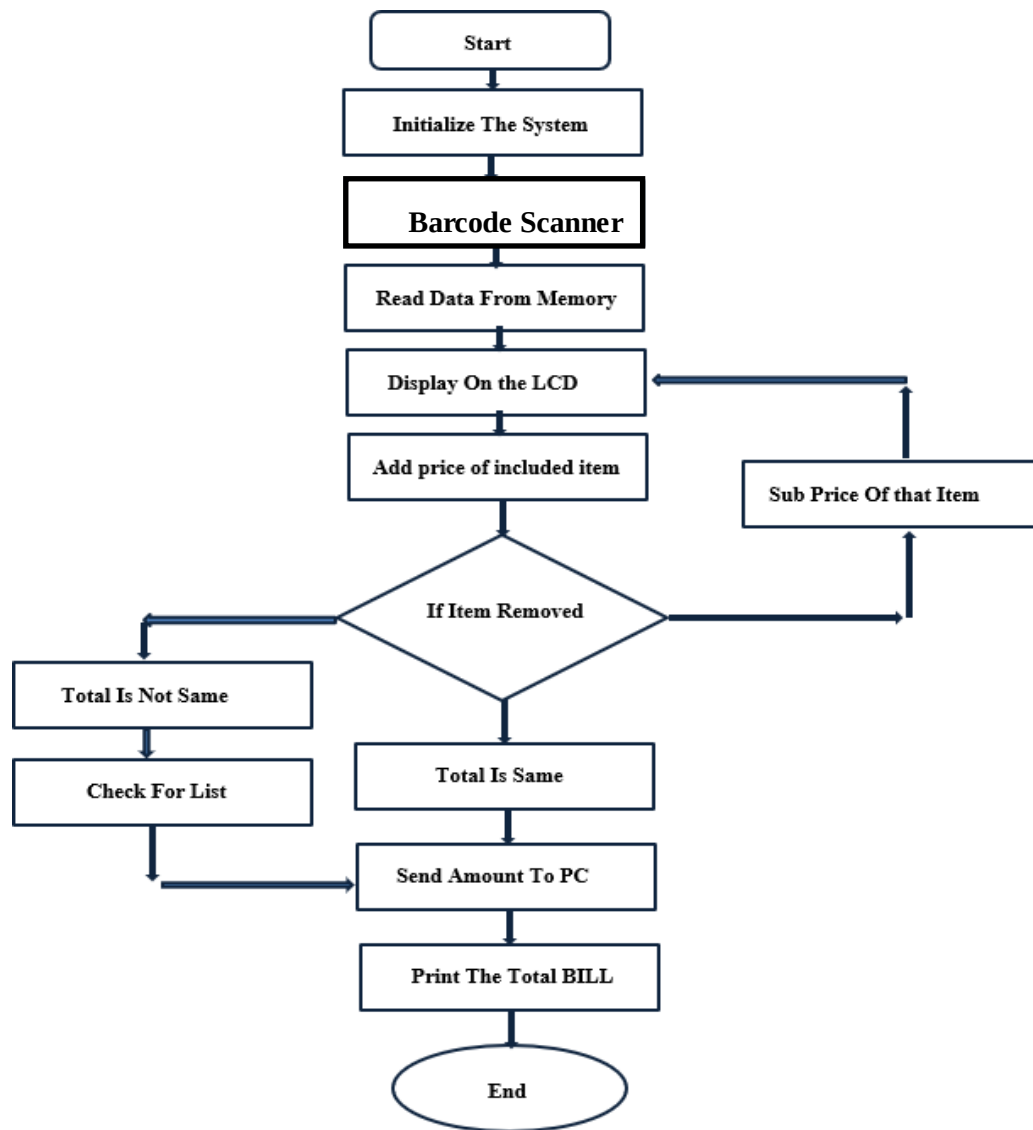


Fig 2-Flow Chart of Smart Billing System

The above **Fig 2** represents the overall flow of the system starting with the initializing the system to the actual bill results been displayed. The project idea focuses on two basic solution variants based on the number of items purchased. Web application for baskets: Aim is to provide billing for less quantity of items that can accommodate in the basket. This application needs to be downloaded and installed on any Android specific device to use full features.

DFD LEVELS:

1. DFD level 0

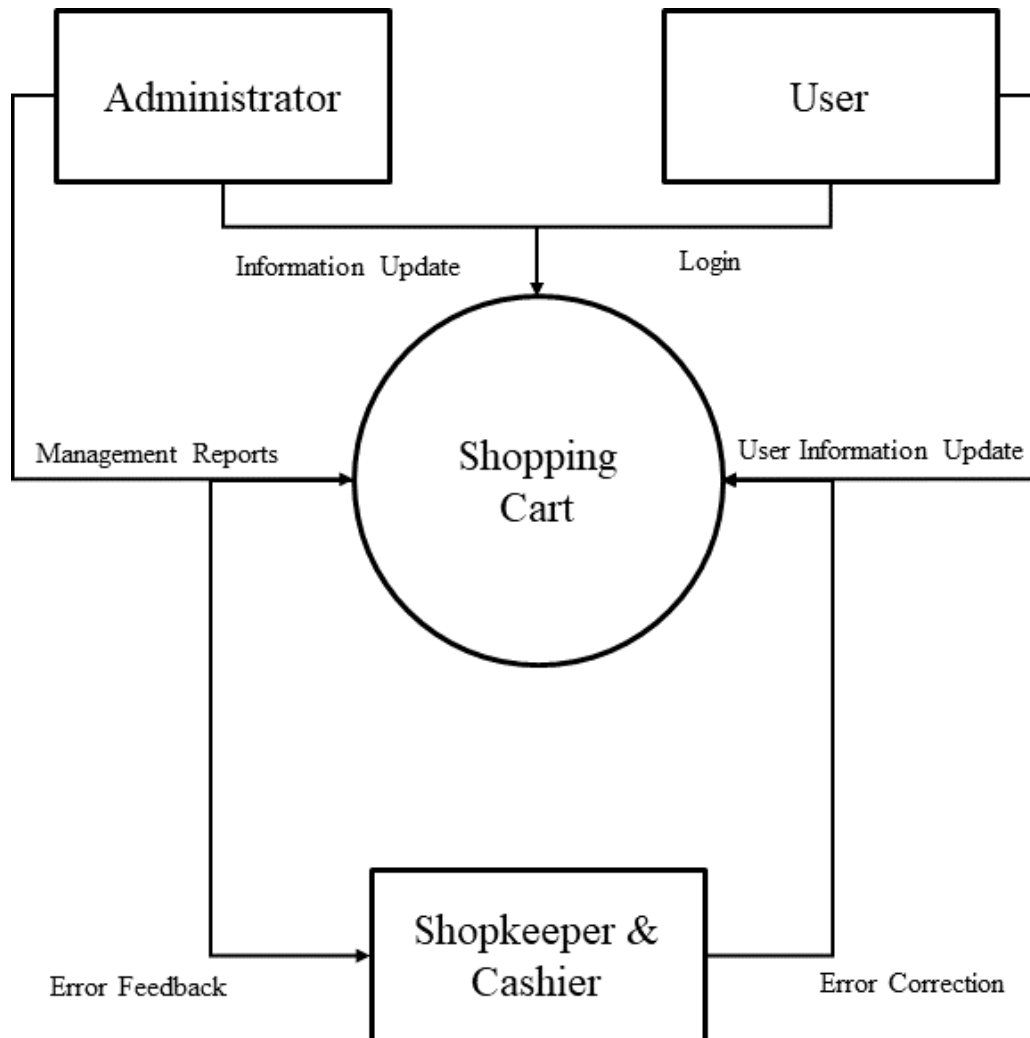
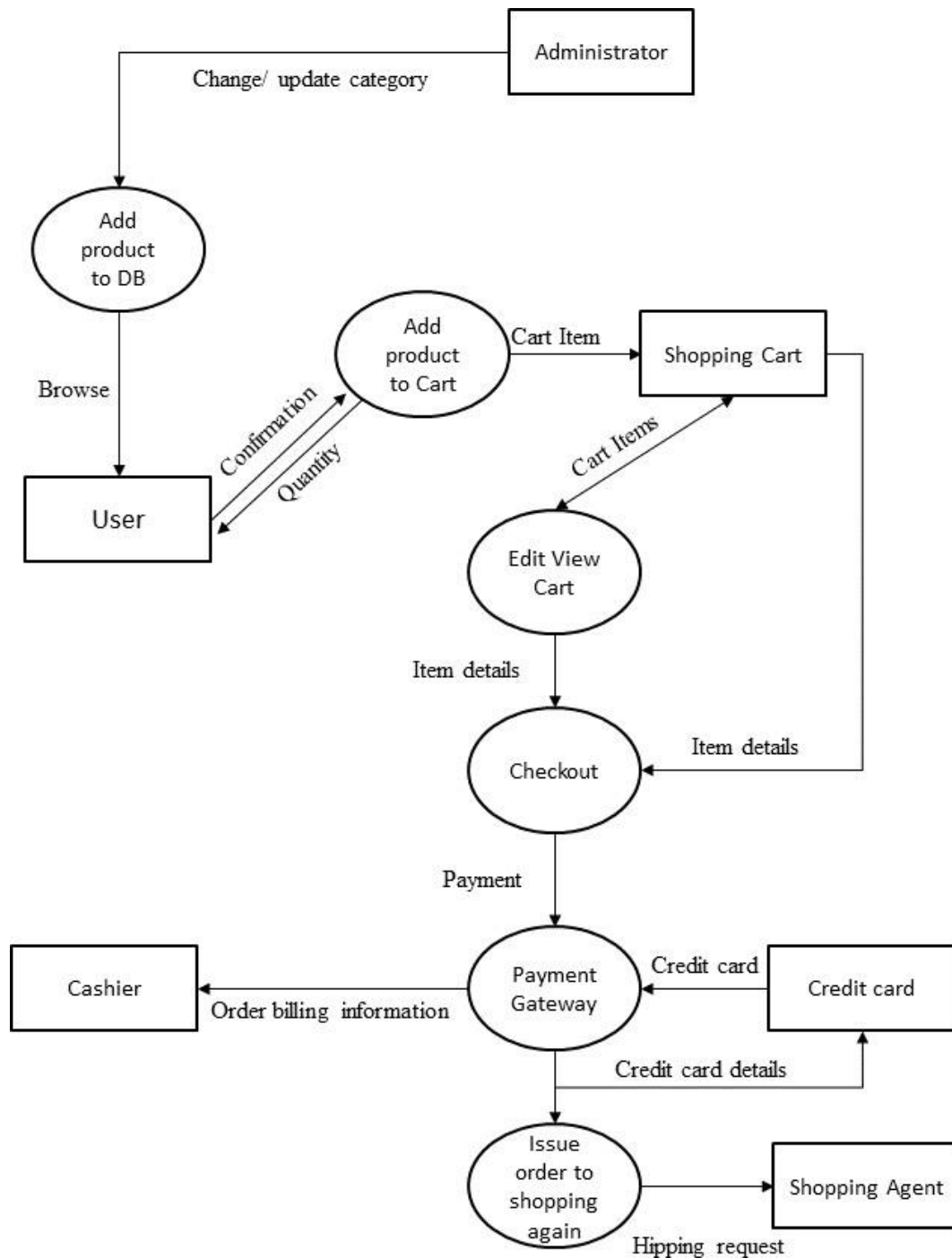


Fig 3-Dataflow diagram level 0

The diagram you sent appears to be a conceptual representation of a shopping cart system, focused on the roles and interactions of the users involved. The diagram depicts a system where users can shop for items, manage their accounts, and provide feedback. Administrators can manage reports, update information, and correct errors.

2.DFD level 1



**Fig 4-Dataflow diagram
level 1**

The diagram illustrates the process of adding a product to a shopping cart and checking out. It starts with a user browsing products. The system then interacts with a payment gateway to process the payment. Once the payment is successful, the user receives a confirmation and their order is sent to the shopping agent, likely to initiate fulfillment. The shopping agent may then send a shipping request.

3.DFD level 2

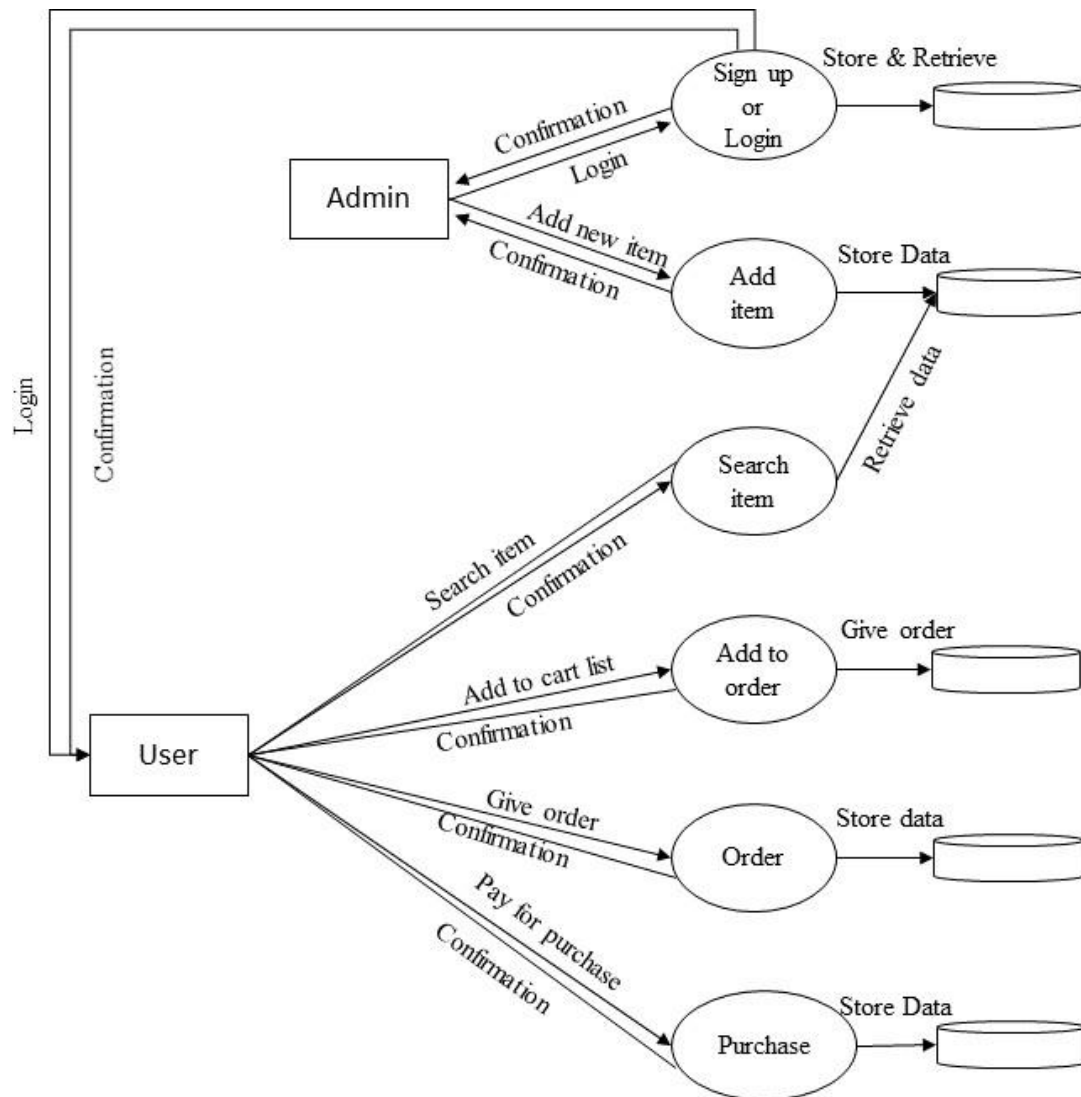


Fig5- Dataflow diagram level 2

The process starts with the user logging in or signing up for an account. Once logged in, the user can browse items and add them to their cart. The admin can also add new items to the system. After selecting the desired items, the user can confirm their order. This triggers a confirmation step for both the user and the admin. The user can then pay for the purchase. Again, confirmation is required from both sides. The process seems to end with another confirmation for both the user and the admin, though it's not entirely clear what this final confirmation signifies. In short, this diagram outlines a shopping system where users browse and purchase items, while admins manage the product database and potentially fulfill orders.

CHAPTER 2

INTRODUCTION

A smart billing system is one that has a scanner and other components that enable it to do a range of activities. Smart billing systems can be utilized in a variety of settings, including retail businesses, to improve user efficiency and convenience. The system can be programmed to track the things being purchased or checked out, calculate the overall cost of the items, and provide personalized recommendations to the user based on previous purchases and preferences. To provide additional functionality and convenience, smart billing systems can be connected with other systems such as databases and payment processing systems.

A smart billing system is a web application that is equipped with barcode scanner, weighing machine, other components that allow it to perform a variety of tasks. System can be used in a variety of environments, such as retail stores, to improve efficiency and convenience for users.

The barcode scanner is another technology that is often used in smart billing systems. It is used to scan the barcodes of objects and transfer the information to the database for processing. The barcode scanner can track the things being purchased or checked out, as well as calculate the total cost or other pertinent information. The Barcode Scanner is used by the web app to scan a product's barcode. You can add a new product to the database or change one that already exists. It can be used to analyses data.

The QR code is a widespread technology used in smart billing systems. The QR Code is a two-dimensional symbol. QR codes are utilized to generate unique basket numbers, allowing us to readily identify the basket on the counter. The QR Code on the baskets is scanned with a smartphone,

Web applications created in programming languages such as PHP can also be used to supplement a smart billing system's functionality. A web application, for example, can retrieve information about the things being purchased or checked out from a database, as well as process payment information and update the database to reflect the user's cart and purchase history.

Finally, a database can be utilized to store and retrieve information about the things being purchased or checked out, as well as the user's cart and previous purchases. The database can be accessed via the web app or via the internet.

A smart billing system is designed to meet these needs by leveraging advanced technology to streamline the billing and payment processes. This

innovative system automates the generation of invoices, minimizes errors, and ensures that billing operations are carried out swiftly and accurately. With a user-friendly interface, the system allows businesses of all sizes to easily manage and customize their billing practices, aligning them with their brand identity and operational needs. Furthermore, it supports real-time updates and multi-channel delivery, ensuring that businesses can keep track of payments and communicate with customers effectively across various platforms. Secure payment integration is another cornerstone of a smart billing system, enhancing the security of financial transactions and facilitating prompt payments. Overall, a smart billing system not only simplifies financial management but also enhances the overall customer experience, reflecting the dynamic nature of modern commerce.

CHAPTER 3

LITERATURE REVIEW

Literature Review

1). In the “**IoT applications on Secure Smart Shopping System**” the **author Ruinian Li, Tianyi Song, Nicholas Capurso, Jiguo Yu, JasonCouture, and Xiuzhen Cheng** implement the idea using UHF RFID reader so every smart cart is equipped with UHF reader. This system consists of microcontroller, LCD, weight sensor, and ZigBee technology. This system is implemented for the security and privacy issues for make the system practical. The final result, final billing can be done in the trolley. So, the users don't have to wait in a queue for long time [1].

2). In the “**The Development of Smart Shopping Cart with the Customer-Oriented Service**” the **author Hsin-Han Chiang, Wan-Ting You, Shu-Hsuan Lin, and Wei-Chih Shih** implement a smart shopping card that can be detected automatically the item is added into the shopping trolley. From the searching of SSC, the navigation the purchasing in the mall is efficiently supplied [2].

3). In the “**Smart Trolley in Mega Mall**” the **author Awati.J.S, S.B.Awati**, They Developed microcontroller based design for users who wait in a queue so avoid the crowd at the billing counter and headache like pulling a trolley. They used LCD display, Max 232, Barcode scanner; RF module, RF transmitter & RF receiver, & Object counter [3]

Sr.NO	PAPER TITLE	YEAR	AUTHORS	ANALYSIS
1	Smart shopping cart	2017	1.Akashay Kumar 2.S Balamurug	This project mainly focused on autonomous technology to scan the products and store it in mobile database and Payment is through mobile banking.
2	Innovative Shopping Cart	2019	1. Prasiddhi K. 2.Dhanasiri Halwai	Using RFID and ZigBee technology, an inexpensive RFID tag can be attached to each product and scan it with RFID reader

3	RFID based Advanced Shopping Trolley for Super Market	2020	1.Manikandan Thiyagarajan, 2. Mohammed Aejaz 3. Nithin Krishna 4. Mohan Kumar	Using RFID technology. RFID reader present in each trolley and RFID tag for each item, reader scans each product rate and brand of the product and the information will be shown on the LCD screen.
4	Smart trolley using Smart Phone and Arduino	2021	1. Harpeet Bedi 2. Kumar S 3. Gupta A	This application uses RFID to scan the products and sends the details to database. Shopping commences by pressing start button and ends by pressing stop button provided in the trolley
5	Smart Trolley	2021	1. Sarmad Ali 2. MahreenRiaz 3. Hugo Femando	In this application consumer can self-scan the barcodes using ultrasonic sensor that is attached to the trolley Items in the trolley with a purchaser ID is linked with supermarket backend database which contains features of the goods such as cost price and other details.
6	RFID-Cloud smart cart system	2021	1. YerlanBerdaliyev 2. AlexPappachen James	Using RFID technology for billing of the purchased item includes particularly calculated PCB, a Wi-Fi, and a power supply to intend the shopping trolley
7	An IOT Based Smart Shopping Cart for Smart Shopping	2021	1. Srinidhi Karjol 2. Anusha K. Holla 3. C. B. Abhilash	This project provides trolley to trolley communication that helps the consumers to shop with friends and family.
8	Smart Cart with Automatic Billing	2022	1. Ankush Yewatkara 2. Faiz Inamdarb	This application provides anti-theft controlling system for a superstore. It will allow online transaction process

			3. Raj Singh 4. Ayushyad	for billing, and it will also give suggestions to the customer for purchasing items.
9	Smart Trolley System For Automated Billing Using RFID And Zigbee	2022	1. P.Chandrasekar 2. T.Sangeetha	The smart trolleys consist of RFID reader LCD display and Zigbee transmitter.

Table no: 1 Literature review

CHAPTER 4

METHODOLOGY

A. Display Screen:

The display screen serves as a visual interface for users to interact with the system. It can display information such as the total cost of items in a retail store or the status of items being checked out in a library. This provides users with real-time feedback and enhances their overall experience.

B. Wireless Communication:

The smart billing system utilizes wireless communication technologies like Wi-Fi or 3G/4G to connect to the internet. This connectivity enables the system to access information and resources online, such as product databases, pricing information, or updates. It enhances the system's versatility and allows for seamless integration with other systems or platforms.

C. Sensors:

Sensors, such as ultrasonic sensors or infrared sensors, are employed to detect and respond to obstacles and changes in the environment. For instance, they can detect when items are placed in the basket or when there are obstructions in the checkout process. This helps streamline operations and ensures smooth functioning of the system.

D. Thermal Printer:

The system utilizes a thermal printer, either direct thermal or thermal transfer, to print generated bill receipts. Thermal printing technology offers fast, efficient, and high-quality printing of receipts. Direct thermal printers use heat to produce images on specially coated paper, while thermal transfer printers use a ribbon to transfer ink onto labels, resulting in permanent prints.

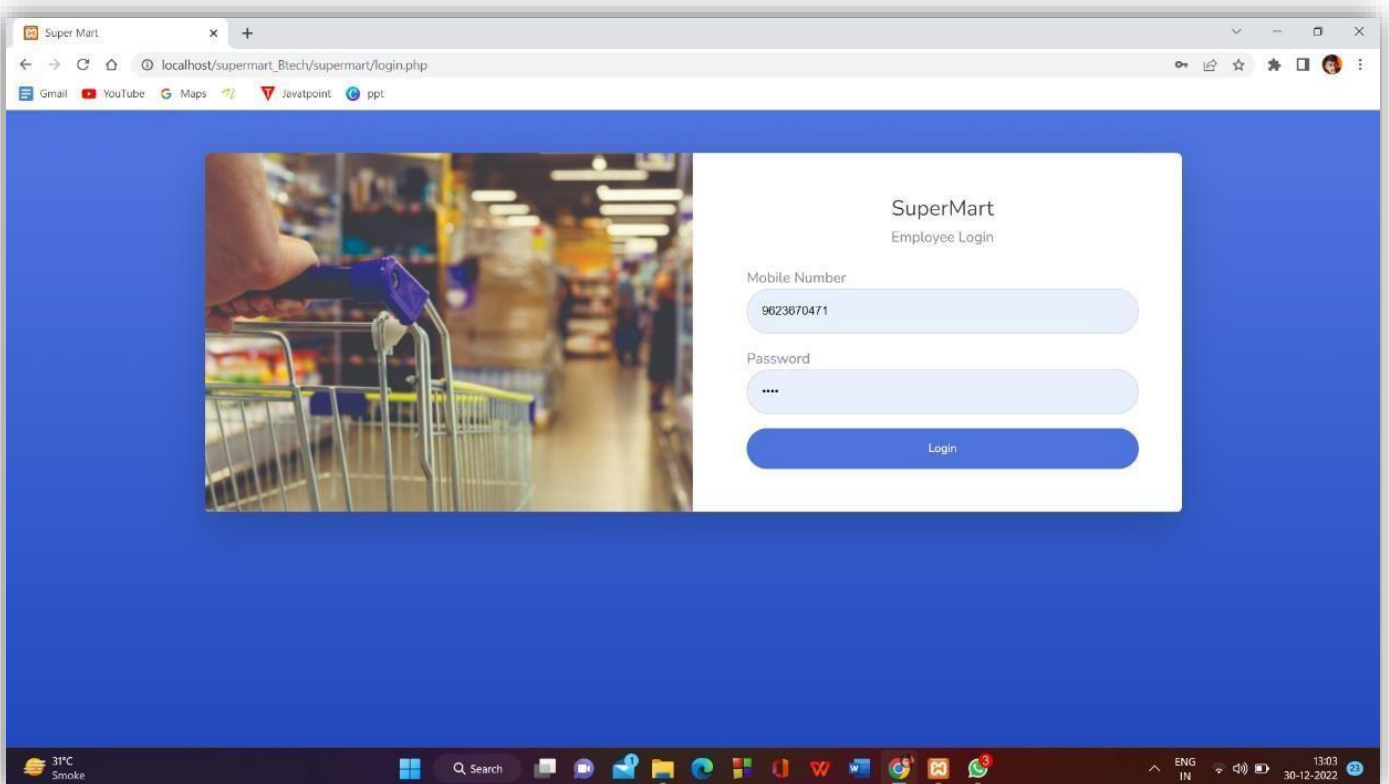
E. Weighing Machine:

To prevent fraud, a weighing machine is integrated into the system and attached to the basket. The calculated weight of items in the database is compared with the actual weight of all purchased products. If there's a match, the items are cleared for billing; otherwise, they are checked manually to ensure accuracy and prevent discrepancies.

F. QR Code:

QR codes are utilized to generate basket numbers, and each basket is assigned a unique QR code. These QR codes are generated using online QR code generators and are attached to each basket. QR codes provide a convenient and efficient way to identify and track baskets throughout the checkout process, facilitating smooth and organized operations.

Screenshots:

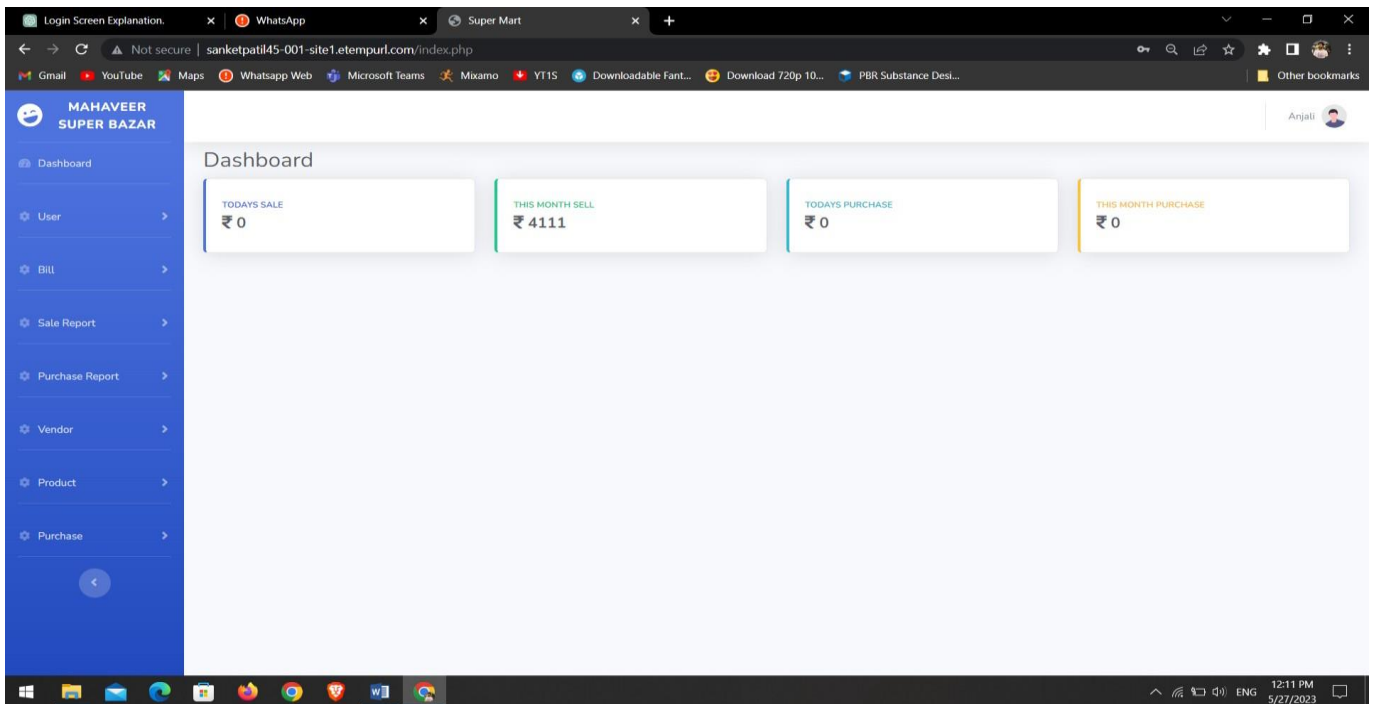


(Screenshot 1) Login Screen:

The login screen, also known as the authentication screen or sign-in screen, is an essential component of many software applications and websites. It serves as the initial point of interaction between the user and the system, where users are required to provide their credentials to gain access to protected features, personalized content, or secure areas.

Key Features:

- i. Error Handling
- ii. Security Measures
- iii. Responsive Design
- iv. Testing and Usability



(Screenshot 2) Dashboard/ Home Screen of Smart Billing System.

The home screen, also known as the main screen or dashboard, is a crucial element of software applications and websites. It serves as the central hub where users land after successfully logging in, providing them with an overview of important information, key functionalities, and navigation options.

Key Features:

1. Data Visualization:

Dashboard is used to display various information like: **Total Sales** for current day, **Monthly sales**, and **Total customers** for current day. It also shows different menu tabs for: **Purchase report, Vendor details, Product details, Purchase details etc.**

2. Usability and User Experience:

Due to availability of details and data on Dashboard screen user experience is always our priority. Users are able to watch synchronized data all over from database to application instantly.

The screenshot shows a web browser window with a sidebar menu on the left containing options: Dashboard, User, Bill, Sale Report, Purchase Report, Vendor, Product, and Purchase. The main content area displays a 'Product Info' form. The form fields are: Product name (text input), GST % (dropdown menu showing 0%), HSN Number (text input), Barcode (text input), Selling Price (text input), Purchase Price (text input), Product Weight(Gm) (text input), Net Weight (text input), and Quantity (text input). A blue 'Submit' button is at the bottom of the form. The browser's address bar shows 'sanketpatil45-001-site1.etimeurl.com/product.php'.

(Screenshot 3) Adding Product Information to the Database:

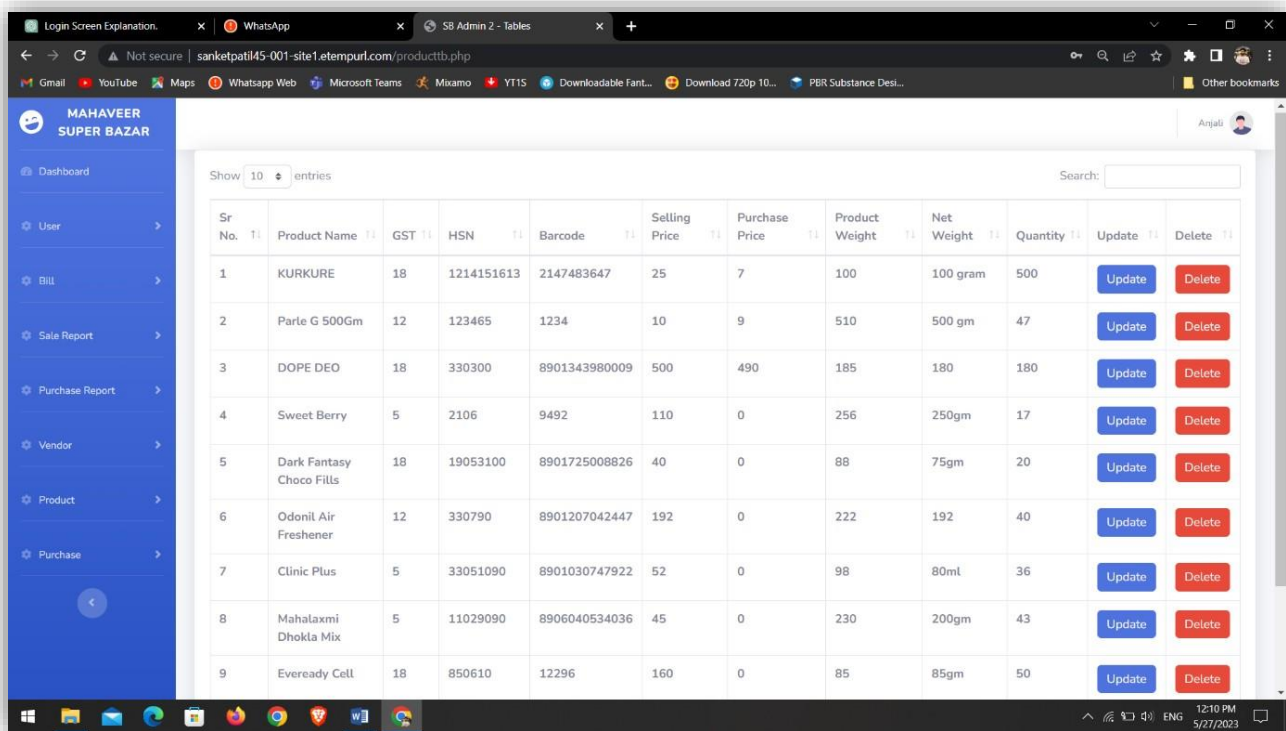
Admin, Shopkeeper can add or remove product. For this process they have separate menu Which adds or removes product strait from database.

i. Adding product:

Simply fill the information as suggested in form i.e. **Product name, GST HSN number, Weight of product, Quantity, Selling price, Manufacturing date, Expiry date.** After which you can **scan barcode** of product to assign unique identification key, and press **Submit** to add in the database.

ii. Remove product:

To remove product head to the option says **REMOVE PRODUCT** and just scan the barcode of product you want to remove and it will be removed from database too.



MAHAVEER SUPER BAZAR

Dashboard

User

Bill

Sale Report

Purchase Report

Vendor

Product

Purchase

Show 10 entries

Search:

Sr No.	Product Name	GST	HSN	Barcode	Selling Price	Purchase Price	Product Weight	Net Weight	Quantity	Update	Delete
1	KURKURE	18	1214151613	2147483647	25	7	100	100 gram	500	Update	Delete
2	Parle G 500Gm	12	123465	1234	10	9	510	500 gm	47	Update	Delete
3	DOPE DEO	18	330300	8901343980009	500	490	185	180	180	Update	Delete
4	Sweet Berry	5	2106	9492	110	0	256	250gm	17	Update	Delete
5	Dark Fantasy Choco Fills	18	19053100	8901725008826	40	0	88	75gm	20	Update	Delete
6	Odonil Air Freshener	12	330790	8901207042447	192	0	222	192	40	Update	Delete
7	Clinic Plus	5	33051090	8901030747922	52	0	98	80ml	36	Update	Delete
8	Mahalaxmi Dhokla Mix	5	11029090	8906040534036	45	0	230	200gm	43	Update	Delete
9	Eveready Cell	18	850610	12296	160	0	85	85gm	50	Update	Delete

(Screenshot 4) Product List:

Product list is shown from the database. It contains all the information about individual product i.e. Name, Quantity available, HSN, Selling price, weight. This form also let Admin or Shopkeeper to edit the product details or remove product from database in just one click.

Mart Billing System

MAHAVEER SUPER BAZAR

Dashboard

User

Bill

Sale Report

Purchase Report

Vendor

Product

Purchase

User Info

First name

Last name

Mobile Number

Birth Date

Role

Address

Submit

(Screenshot 5) Adding User to the Database:

Admin has the authority to add or remove the users like Cashier, Vendor or Shopkeeper.

MAHAVEER SUPER BAZAR

Dashboard

User

Bill

Sale Report

Purchase Report

Vendor

Product

Purchase

Show 10 entries

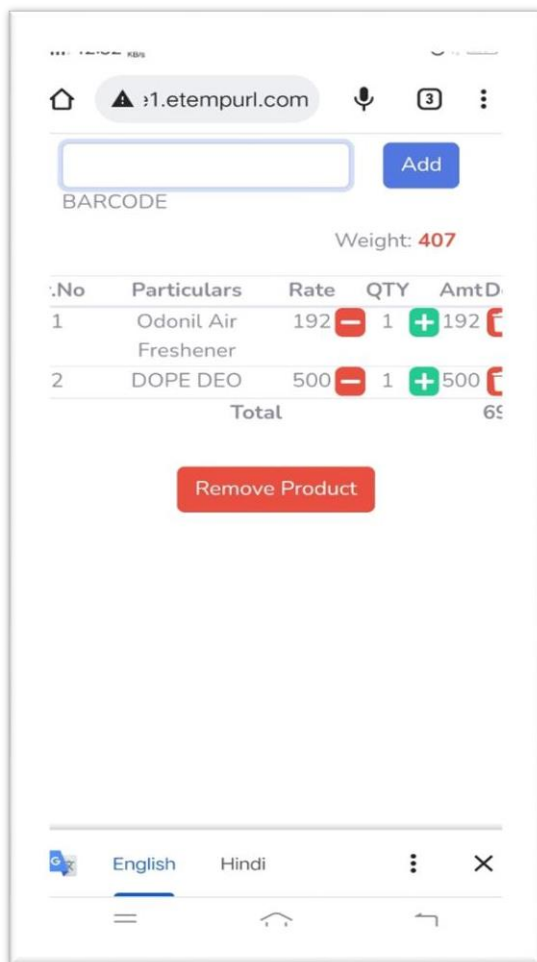
Search:

id	First Name	Last Name	Mobile Number	Birth Date	Role	Address	Update	Delete
10	Sanket	Patil	1234567890	2022-12-15	ShopKeeper	kolhapur	Update	Delete
13	Anjali	Yadav	7517303546	2000-01-01	Admin	Chinchwad	Update	Delete
14	Guruprasad	Desai	9623670471	2022-12-20	Cashier	kolhapur	Update	Delete

Showing 1 to 3 of 3 entries

Previous 1 Next

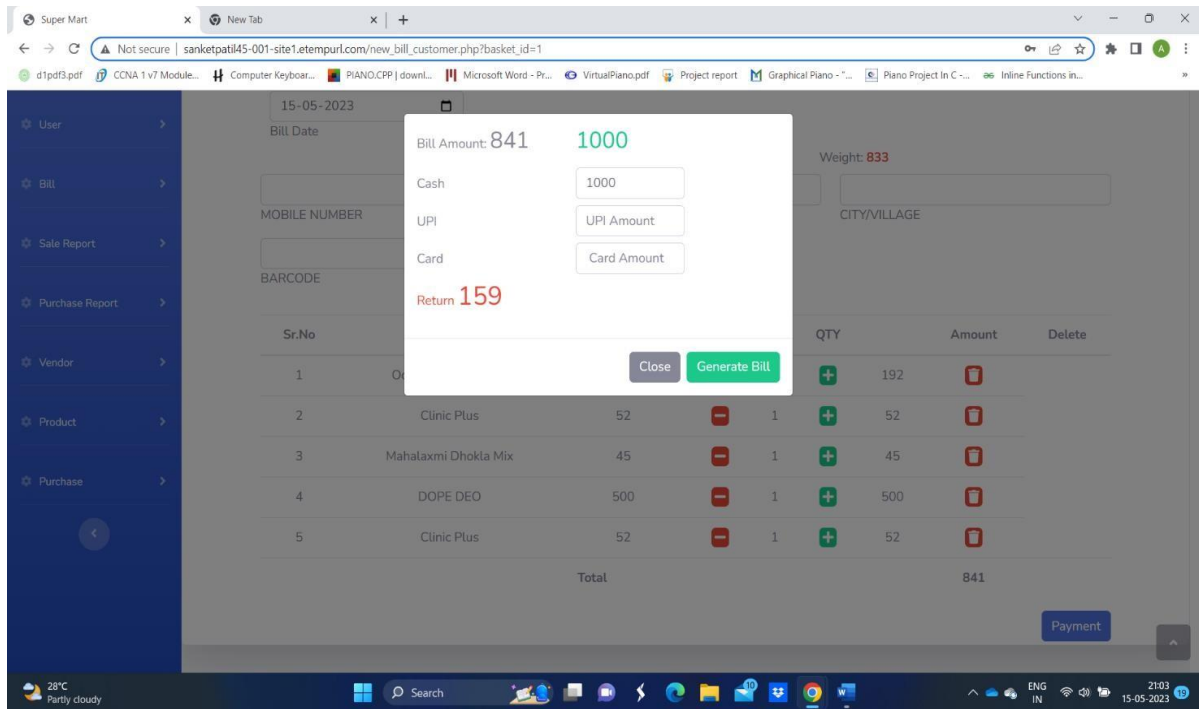
(Screenshot 6) User List: It displays all users of the system.



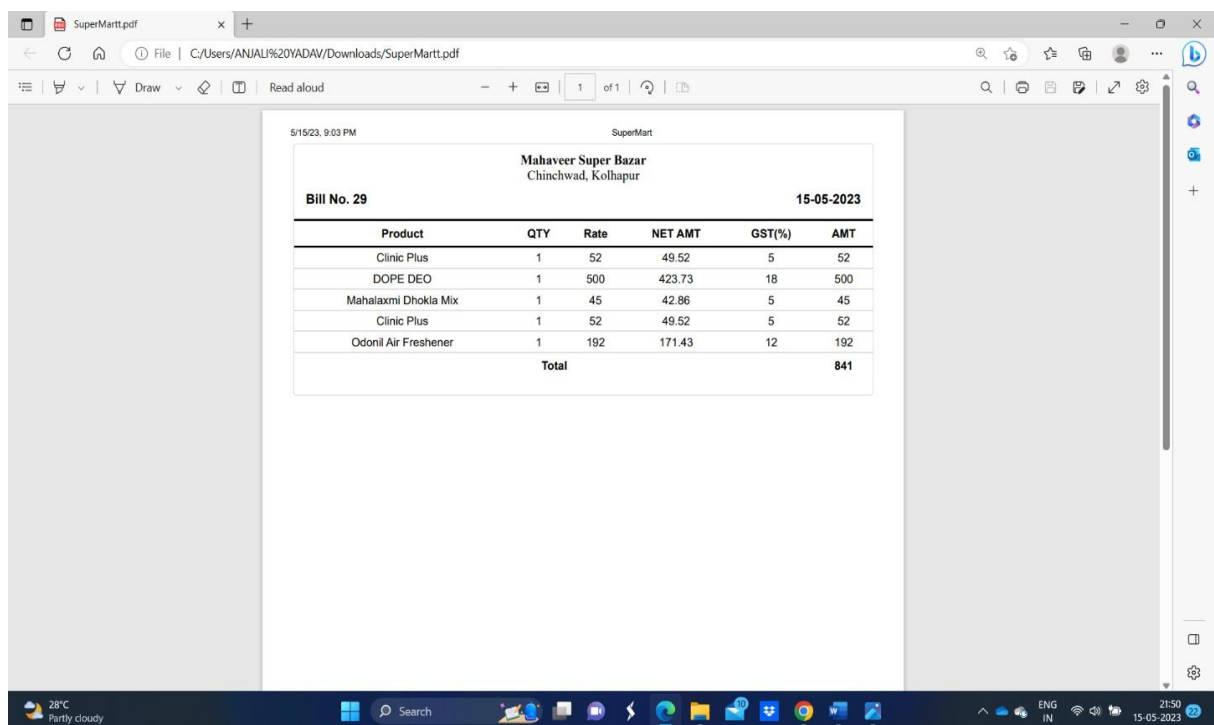
(Screenshot 7) Add and remove items on display screen (customer end):

Customers who are shopping have their smartphones, they have to scan the QR code provided on the basket. After scanning QR code they will be redirected to the web-page which will look like screenshot 11. In this customers can edit the products they have already purchased. Either customer can add more products or remove existing ones.

Mart Billing System



(Screenshot 8) Payment Options: Choose variety of payment options to pay bill amount. If cash paid by customer is more than bill amount, then it will display the return cash to be paid.



(Screenshot 9) Generate the final Bill: Final bill print is generated, with all purchased productlist, amount of products and cash details.

CHAPTER 5

TESTING RESULT

Introduction

In There are several testing mechanisms that can be used to ensure the functionality and reliability of a smart billing system. Some examples include:

1. **Unit testing:** Unit testing involves testing individual components or units of the smart billing system separately to verify their functionality. Unit testing is a fundamental software testing approach where individual components or units of a software system, such as a smart billing system, are tested in isolation from the rest of the system. The primary goal of unit testing is to validate that each unit performs as designed. A unit typically refers to the smallest testable part of the application, which can be a function, method, class, or module, depending on the programming language and the design of the software.

Test number	Test description	Test input	Expected output	Actual output	Result
1	Add product functionality	User input: "Add new product"	Product is added to list	Product is added to list	Pass
2	Daily purchase report functionality	User input: "Generate daily purchase report"	Daily purchase report is generated	Daily purchase report is not generated	Fail
3	Product-wise purchase report functionality	User input: "Generate product-wise purchase report"	Product-wise purchase report is generated	Product-wise purchase report is generated	Pass
4	Add vendor functionality	User input: "Add new vendor"	Vendor is added to list	Vendor is added to list	Pass
5	Vendor list functionality	User input: "View vendor list"	List of vendors is displayed	List of vendors is displayed	Pass
6	Purchase report functionality	User input: "Generate purchase report"	Purchase report is generated	Purchase report is not generated	Fail
7	Product list functionality	User input: "View product list"	List of products is displayed	List of products	Pass

				is displayed	
8	New purchase functionality	User input: "Add items to purchase list"	Items are added to purchase list	Items are added to purchase list	Pass
9	Purchase list functionality	User input: "View purchase list"	List of items in purchase is displayed	List of items in purchase is displayed	Pass
10	Bill functionality	User input: "Generate bill"	Bill is generated and displayed	Bill is generated and displayed	Pass

2. Integration testing: Integration testing involves testing the integration of different components and systems within the smart billing system to ensure that they are working together correctly. Integration testing is a level of software testing where individual units or components of a software application are combined and tested as a group. The primary goal of integration testing is to identify any discrepancies, defects, or failures in the interaction between integrated units or systems within the application. In essence, while unit testing ensures that each part of the system works correctly on its own, integration testing ensures that they work together correctly.

Test number	Test description	Test input	Expected output	Actual output	Result
1	Integration test - Barcode scanner	User scans barcode of item	Item is added to cart	Item is added to cart	Pass
2	Integration test - Weight machine	User adds item to cart	Weight of item is displayed and added to total	Weight of item is displayed and added to total	Pass
3	Integration test - Web application	User accesses cart through web app	Cart is displayed and can be modified	Cart is displayed and can be modified	Pass
4	Integration test - Database	User adds item to cart	Item is added to cart and database	Item is added to cart and database	Pass

5	Integration test - Barcode scanner and weight machine	User scans barcode and adds item to cart	Item is added to cart and weight is displayed	Item is not added to cart and weight is not displayed	Pass
6	Integration test - Web application and database	User adds item to cart through web app	Item is added to cart and database	Item is not added to cart and database	pass

3. System testing: System testing is a critical phase in the software development lifecycle where the entire application, in this case, the smart billing system, is tested to verify that it functions according to the specified requirements. This type of testing is performed after integration testing, where individual units or modules are combined and tested. System testing aims to evaluate the system's compliance with its specified requirements and to ensure it is free from defects.

Test number	Test description	Test input	Expected output	Actual output	Result
1	Verify that the smart billing system is able to process orders and generate bills correctly	User adds items to cart and checks out	Bill is generated correctly	Bill is generated correctly	Pass
2	Verify that the smart billing system is able to correctly update inventory levels after a purchase	User adds items to cart and checks out	Inventory levels are updated correctly	Inventory levels are updated correctly	Pass
3	Verify that the smart billing system is able to	Multiple users adding items to	System processes orders without	System processes orders without	Pass

	handle high workloads without issues	cart and checking out simultaneously	delays or errors	delays or errors	
4	Verify that the smart billing system is able to function correctly in different environments	Testing in different lighting and temperature conditions	cart functions correctly in all environments	cart functions correctly in all environments	Pass
5	Verify that the smart billing system is able to handle communication with other systems, such as a server or database	User adds items to cart and checks out, with communication with server	Smart billing system communicates with server correctly	Smart billing system communicates with server correctly	Pass

CHAPTER 6

CONCLUSION

A Smart Billing System using a barcode scanner, printer, QR code, weighing machine and sensors can be a highly effective and efficient tool for a variety of applications.

The use of QR code allows for a high degree of customization and control over the cart's functions and features, and the barcode scanner enable the cart to interact with its environment and gather relevant information.

Overall, the implementation and testing of a smart cart using these technologies has been successful, and the cart and billing system has demonstrated good performance and reliability.

Future Work:

However, there is always room for improvement and further work. Some potential areas for further development and enhancement include:

a. Adding additional sensors or functionality:

Additional sensors or functionality, such as a camera or a voice recognition system, could be added to the cart to enhance its capabilities.

b. Improving the user interface:

The user interface could be further refined and optimized to make it more intuitive and user-friendly.

c. Adding the range and connectivity of the Bluetooth and GPS:

The range and connectivity of the Bluetooth and GPS could be increased to allow the cart to operate more efficiently and effectively.

d. Adding RFID capability:

Adding RFID capability to the smart cart could enable more advanced tracking and identification functionality.

Overall, there are many opportunities for further development and improvement of a smart cart using Arduino, a barcode scanner and with continued work and innovation, the cart can be further enhanced to meet the evolving needs and requirements of users.

CHAPTER 7

REFERENCES

1. Dr. Suryaprasad J, Praveen Kumar B. O, Roopa D. Arjun - “A Novel Low-Cost Intelligent Shopping Cart”, Proceedings of the 2nd IEEE International Conference on Networked Embedded Systems for Enterprise Applications, NESEA 2011, Perth, Australia, December 89, 2011.
2. Suganya R Swarnavali. N, Vismitha. S, Mrs G.M. Rajathi “Automated Smart Trolley with Smart Billing Using Arduino” in International Journal of Advanced Research in Computer Engineering & Technology (IJARCET) Volume 4 Issue III, March 2016.
3. Zeeshan Ali, Prof. Reena Sonkusare, “RFID Based Smart Shopping and Billing”, International Journal of Advanced Research in Computer and Communication Engineering, Volume. 2, Issue 12, December 2013
4. Anwar, S., & Simanjuntak, R. T. (2021). Moderating Effect Of The Internet Understanding On The Relationship Between The E-Filing And E-Billing With Individual Taxpayer Compliance.
5. Ayem, S., & Wahidah, U. (2022). Faktor-Faktor Yang Mempengaruhi Pemanfaatan E-Billing Pajak Dengan Menggunakan Technology Acceptance Model (Tam) Pada Umkm Di Daerah Istimewa Yogyakarta.
6. Bandiyono, A., & Choirun, H. M. (2019). Services E-Filing And E-Billing To Increase Tax Compliance And Acceptance.
7. Dwitrayani, M. C. (2020). Pengaruh Penerapan E-Billing System, E-Filing System Dan Penurunan Tarif Pajak Penghasilan Final Terhadap Kepatuhan Wajib Pajak Pelaku Umkm Di Kota Denpasar.
8. Lailiyah, N., Karolina, L., & Sebayang, B. (2019). Effectiveness Of E-Billing System In Tax Payments For Taxpayers.
9. Alan Dix, Janet Finlay, Gregory Abowd, and Russell Beale (2003): Human–Computer Interaction. 3rd Edition. Prentice Hall, 2003. <http://hcibook.com/e3/> ISBN 0-13-046109-1

CHAPTER 8

SAMARTH KRUPA TRADERS

S.S.Teli-8975767018
S.A.Teli-9527952812

Date-15/09/2023.

Myself, Suresh Shivappa Teli owner of Samarth Krupa Traders accept your sponsorship request for your mega project on subject Billing system for our firm.

I am glad to give a sponsorship for your excellence thought of thinking. I hope the project will be best as our needs by your guidance.

Thank-You.

समर्थ कृपा ट्रेडर्स


मालिक



Mart Billing System

D. A. Nikam, Sangram B. Patil, Tanmay R. Patil, Sushant S. Teli, Tanmay T. Magdum

1 Deepali Nikam Dr. J.J. Magdum College of Engineering, Jaysingpur.

2 Sangram Patil, Dr. J.J. Magdum College of Engineering, Jaysingpur.

3 Tanmay Patil, Dr. J.J. Magdum College of Engineering, Jaysingpur.

4 Sushant Teli Dr. J.J. Magdum College of Engineering, Jaysingpur.

5 Tanmay Magdum, Dr. J.J. Magdum College of Engineering, Jaysingpur.

Abstract

The Mart Billing System for Mega Project is a comprehensive software solution designed to streamline and automate the billing process for large-scale mart operations. In the contemporary retail landscape, efficient billing systems are indispensable for managing transactions, inventory, and customer interactions seamlessly. This project addresses the complexities of managing billing operations in mega marts by integrating advanced features and technologies.

Key features of the Mart Billing System include a user-friendly interface for both cashiers and customers, robust inventory management capabilities, real-time transaction processing, and secure payment options. The system is designed to handle high volumes of transactions efficiently, ensuring minimal wait times for customers during peak hours. Moreover, the Mart Billing System incorporates advanced reporting and analytics tools, enabling mart managers to gain valuable insights into sales trends, inventory turnover rates, and customer preferences. These insights empower decision-makers to optimize pricing strategies, stock levels, and promotional campaigns for enhanced profitability and customer satisfaction.

INTRODUCTION

In the dynamic landscape of modern retail, the efficient management of transactions stands as a cornerstone for success. Mega projects encompassing large-scale marts, supermarkets, or hypermarkets serve as bustling hubs of commerce, catering to diverse consumer needs. However, amidst the vast array of products and the flurry of transactions, the conventional billing systems often encounter limitations in handling the complexity and scale of operations.

To address these challenges and propel the retail experience into the realm of efficiency and innovation, the development of a robust Mart Billing System emerges as a pivotal endeavor. This paper explores the design, implementation, and implications of such a system tailored specifically for mega projects.

The Mart Billing System envisioned herein is not merely a technological upgrade but a holistic solution engineered to streamline operations, enhance customer satisfaction, and empower management with insightful analytics. By integrating cutting-edge technologies such as artificial intelligence, data analytics, and seamless connectivity, this

system promises to revolutionize the retail landscape.

This paper delves into the fundamental components of the Mart Billing System, elucidating its architecture, functionalities, and the transformative potential it holds for mega projects. Furthermore, it examines the strategic implications of adopting such a system, including its impact on operational efficiency, customer engagement, and competitive positioning within the market.

Through a comprehensive analysis, this paper aims to underscore the significance of embracing innovation in retail management, particularly in the context of mega projects. By embracing the Mart Billing System, stakeholders can unlock new avenues for growth, differentiation, and sustainable success in an increasingly dynamic and competitive retail environment.

Literature Review:

The literature review presents a diverse range of studies focusing on the development and implementation of smart shopping cart systems aimed at enhancing the shopping experience and streamlining the billing process. In 2017, Kumar and Balamurugan introduced a Smart Shopping Cart that leveraged autonomous technology for product scanning and mobile banking for payment. Building on this concept. [1]

Prasiddhi K. and Dhanasiri Halwai in 2019 proposed an Innovative Shopping Cart utilizing RFID and ZigBee technology to facilitate product scanning and data transmission. [2]

In 2020, Thiagarajan, Aejaz, Krishna, and Kumar introduced an RFID-based Advanced Shopping Trolley for Supermarkets, employing RFID technology for product scanning and displaying product information on an LCD screen. Subsequently, in 2021 [3], Bedi, S, Gupta A, Ali, Riaz, and Fernando presented Smart Trolley systems utilizing RFID and ultrasonic sensors for product scanning and database integration for item details and pricing.

Continuing the trend, Berdaliyev, James, Karjol, Holla, Abhilash, Yewatkara, Inamdarb, Singh, Ayushyad, and Chandrasekar in 2021 and 2022 explored RFID technology further, incorporating features such as Wi-Fi connectivity, automatic billing, anti-theft measures, and trolley-to-trolley communication, thereby enhancing shopping convenience and security. [4]

Paper Studies	Work	Dataset used	Method	Results
Smart shopping cart	Smart Shopping Cart technology, aiming to enhance the efficiency and convenience of billing processes in retail environments.	Additional data fields or parameters	Automated algorithms are used to generate invoices based on predefined criteria such as product categories.	This project mainly focused on automating technology to scan the products and store it in a mobile database and Payment is through mobile banking.
Innovative Shopping Cart	Aiming to enhance user experience and streamline the billing process by incorporating features such as personalized recommendations.	Customer purchase history and user behavior analytics to optimize and streamline billing processes.	User-friendly shopping cart interfaces, personalized recommendations, secure payment gateways.	Using RFID and ZigBee technology, an inexpensive RFID tag can be attached to each product and scan it with RFID reader.
RFID based Advanced Shopping Trolley for Super Market	Focusing on enhancing checkout processes, inventory management, and customer experience within the Mart Billing System.	Capturing RFID-tagged product information, transaction records.	Advanced shopping trolleys to enable automatic item identification, real-time inventory tracking.	Using RFID technology, RFID reader present in each trolley and RFID tag for each item, reader scans each product rate.

Additionally, the integration of ZigBee transmitters in the Smart Trolley System for Automated Billing, proposed by Chandrasekar and Sangeetha in 2022[5], further expands the capabilities of smart shopping carts in facilitating automated billing processes.

Overall, these studies demonstrate the continuous evolution and innovation in smart shopping cart systems, driven by advancements in RFID, sensor technology, wireless communication, and database integration, [6] with a common goal of optimizing the shopping experience and improving billing efficiency in retail environments.

The literature review for the Mart Billing System for mega projects provides a comprehensive overview of existing research and practices related to billing management in large-scale construction projects. Numerous studies have highlighted the challenges associated with traditional billing methods, including manual data entry errors, delays in invoice processing, and difficulties in tracking project expenses. [7]

Scholars have emphasized the importance of adopting technology-driven solutions to address these challenges effectively. [8] Various billing systems and software applications have been developed for the construction industry, offering features such as automated invoice generation, real-time expense tracking, and integration with accounting systems.

Human-Computer Interaction (HCI) is "a discipline concerned with the design, evaluation and implementation of interactive computing systems for human use and with the study of major phenomena surrounding them." [9]

II. Methodology

The methodology for the Mart Billing System paper entails a systematic approach encompassing several pivotal steps crucial for its design, development, and evaluation. Initially, a meticulous Requirement Analysis is conducted to delineate the specific needs of the billing system for mega projects. This involves gathering insights through interviews, surveys, and an in-depth examination of prevailing billing practices. Key functionalities, including automated invoice generation, expense tracking, and integration with accounting software, are discerned during this phase. Subsequently, a comprehensive Literature Review is undertaken to scrutinize existing literature on billing systems, construction project management, and related technologies. Drawing from previous research papers and case studies, methodologies, technologies, and frameworks adaptable to the Mart Billing System are identified.

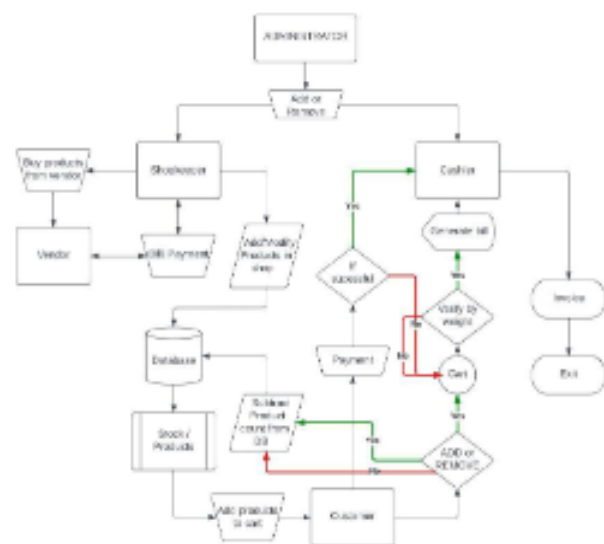


Fig 1: Architecture of Proposed system

Following this, the System Design phase is embarked upon, wherein the architecture and components of the Mart Billing System are meticulously delineated based on the amassed requirements and insights gleaned from the literature review. The database structure, user interfaces, modules, and integration points with other systems are meticulously defined during this phase. Additionally, the suitable technology stack is determined, factoring in considerations such as scalability, security, and compatibility with existing infrastructure.

Upon completion of the design phase, the Development phase commences, wherein the designed system is brought to fruition using appropriate programming languages, frameworks, and tools. Modules for automated invoice generation, expense tracking, real-time data integration, and reporting functionalities are developed with careful attention to detail. Furthermore,

security features such as user authentication, data encryption, and access control mechanisms are integrated to safeguard sensitive financial information.

Once the development phase is concluded, the system undergoes rigorous Testing to ascertain its functionality, reliability, and performance. Unit testing, integration testing, and system testing are conducted to identify and rectify any bugs or issues. Validation against defined requirements and use cases ensures alignment with stakeholders' expectations.



Fig 2: DFD Level 0

III. Propose System

The proposed Mart Billing System aims to revolutionize the billing process within the context of mega projects, offering a comprehensive solution tailored specifically to address the unique challenges encountered in large-scale construction endeavors. Leveraging insights from existing literature and drawing upon innovative technologies, the system integrates advanced features to streamline billing activities and enhance overall project management efficiency.

Central to the proposed system is the implementation of autonomous technology for seamless product scanning and data acquisition, reducing reliance on manual processes and minimizing errors. Through the utilization of RFID technology embedded within smart shopping carts or trolleys, the system enables automatic identification of project-related items, facilitating real-time tracking of expenses and inventory management.

Furthermore, the Mart Billing System incorporates mobile banking capabilities, allowing for secure and convenient payment transactions directly through mobile devices. By integrating with accounting software, the system ensures accurate and timely invoicing, while customizable reporting functionalities provide project managers with valuable insights into expenditure patterns, enabling informed decision-

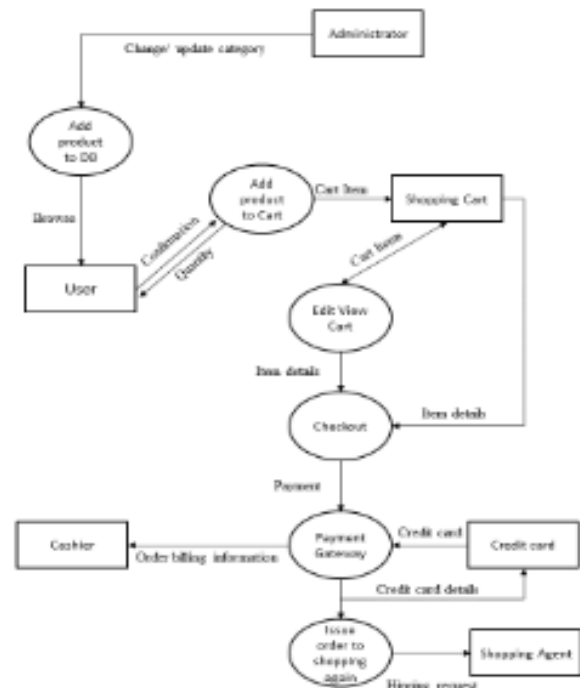


Fig 3: DFD Level 1

A level 1 data flow diagram (DFD) for a mart billing system would show the main processes and data stores involved in the system. The processes would be represented by rectangles, the data stores by cylinders, and the data flows by arrows. The data flows would show how data moves between the processes and data stores.

The data flows between the processes and data stores would show how data moves through the system. For example, the customer data would flow from the customer process to the inventory process and the payment process. The inventory data would flow from the inventory process to the customer process and the payment process. The payment data would flow from the payment process to the customer process. It also identifies intimal data stores of Login, Products, Payment, Sales, purchasing that must be present in order for the Super Market system to do its job, and shows the flow of data between the various parts of Stores, Purchasing, Products, Login, Payment of the system. DFD Level 1 provides a more detailed breakout of pieces of the 1st level DFD. You will highlight the main functionalities of Super Market

The level 1 DFD would provide a general overview of the mart billing system. More detailed DFDs could be created to show the specific steps involved in each process.



IV. Future Work:

1. Integration with Emerging Technologies: Explore the integration of emerging technologies such as blockchain and artificial intelligence to further enhance the capabilities of the Mart Billing System.

2. Enhanced Data Analytics: Implement advanced data analytics techniques to extract valuable insights from the billing system data.

3. Mobile Application Development: Develop a dedicated mobile application to complement the Mart Billing System, providing users with convenient access to billing information, real-time updates, and payment functionalities on their smartphones or tablets.

4. Expansion to Multi-Project Management: Extend the capabilities of the Mart Billing System to support multi-project management scenarios, allowing users to seamlessly manage billing activities across multiple concurrent projects.

5. Enhanced Security Measures: Implement additional security measures to safeguard sensitive financial data and protect against potential cyber threats.

6. Customization and Scalability: Enhance the customization options and scalability of the Mart Billing System to accommodate the diverse needs of different mega projects. This could involve developing modular components that can be easily configured and scaled up or down based on project requirements.

7. User Training and Support: Offer comprehensive user training programs and ongoing technical support to ensure effective adoption and utilization of the Mart Billing System. This could involve developing user guides, tutorials, and online resources.

8. Feedback Mechanisms and Continuous Improvement: Establish feedback mechanisms to gather input from users and stakeholders regarding their experiences with the Mart Billing System. Use this feedback to identify areas for improvement and implement iterative enhancements to the system over time, ensuring that it remains aligned with evolving industry trends and user needs.

V. Conclusion:

The Mart Billing System presents a pioneering solution tailored to the complexities of mega projects, promising to streamline billing processes, enhance accuracy, and ultimately contribute to the success and profitability of large-scale construction endeavors.

VI.

References:

1. Dr. Suryaprasad J, Praveen Kumar B. O, Roopa D. Arjun - "A Novel Low-Cost Intelligent Shopping Cart", Proceedings of the 2nd IEEE International Conference on Networked Embedded Systems for Enterprise Applications, NESEA 2011, Perth, Australia, December 89, 2011.

2. Suganya R. Swarnavali. N, Vismitha. S, Mrs G.M. Rajathi "Automated Smart Trolley with Smart Billing Using Arduino" in International Journal of Advanced Research in Computer Engineering & Technology (IJARCET) Volume 4 Issue III, March 2016.

3. Zeeshan Ali, Prof. Reena Sonkusare, "RFID Based Smart Shopping and Billing", International Journal of Advanced Research in Computer and Communication Engineering, Volume. 2, Issue 12, December 2013

4. Anwar, S., & Simanjuntak, R. T. (2021). Moderating Effect Of The Internet Understanding On The Relationship Between The E-Filing And E-Billing With Individual Taxpayer Compliance.

5. Ayem, S., & Wahidah, U. (2022). Faktor-Faktor Yang Mempengaruhi Pemanfaatan E-Billing Pajak Dengan Menggunakan Technology Acceptance Model (Tam) Pada Umkm Di Daerah Istimewa Yogyakarta.

6. Bandiyono, A., & Choirun, H. M. (2019). Services E-Filing And E-Billing To Increase Tax Compliance And Acceptance.

7. Dwitrayani, M. C. (2020). Pengaruh Penerapan E-Billing System, E-Filing System Dan Penurunan Tarif Pajak Penghasilan Final Terhadap Kepatuhan Wajib Pajak Pelaku Umkm Di Kota Denpasar.

8. Lailiyah, N., Karolina, L., & Sebayang, B. (2019). Effectiveness Of E-Billing System In Tax Payments For Taxpayers.

9. Alan Dix, Janet Finlay, Gregory Abowd, and Russell Beale (2003): Human-Computer Interaction. 3rd Edition. Prentice Hall, 2003. <http://hcibook.com/e3/> ISBN 0-13-046109-1

DOI: 10.55041/IJSREM30242



ISSN: 2582-3930

Impact Factor: 8.448

INTERNATIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING & MANAGEMENT

An Open Access Scholarly Journal || Index in major Databases & Metadata

CERTIFICATE OF PUBLICATION

International Journal of Scientific Research in Engineering & Management is hereby awarding this certificate to

Sangram B. Patil

in recognition to the publication of paper titled

Mart Billing System

published in IJSREM Journal on **Volume 08 Issue 04 April, 2024**

www.ijsrem.com


Editor-in-Chief
IJSREM Journal

ijsremjournal@gmail.com

DOI: 10.55041/IJSREM30242



ISSN: 2582-3930

Impact Factor: 8.448

INTERNATIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING & MANAGEMENT

An Open Access Scholarly Journal || Index in major Databases & Metadata

CERTIFICATE OF PUBLICATION

International Journal of Scientific Research in Engineering & Management is hereby awarding this certificate to

Tanmay R. Patil

in recognition to the publication of paper titled

Mart Billing System

published in IJSREM Journal on **Volume 08 Issue 04 April, 2024**

www.ijsrem.com


Editor-in-Chief
IJSREM Journal

ijsremjournal@gmail.com

DOI: 10.55041/IJSREM30242



ISSN: 2582-3930

Impact Factor: 8.448

INTERNATIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING & MANAGEMENT

An Open Access Scholarly Journal || Index in major Databases & Metadata

CERTIFICATE OF PUBLICATION

International Journal of Scientific Research in Engineering & Management is hereby awarding this certificate to

Sushant S. Teli

in recognition to the publication of paper titled

Mart Billing System

published in IJSREM Journal on **Volume 08 Issue 04 April, 2024**

Editor-in-Chief
IJSREM Journal

www.ijsrem.com

ijsremjournal@gmail.com

DOI: 10.55041/IJSREM30242



ISSN: 2582-3930

Impact Factor: 8.448

INTERNATIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING & MANAGEMENT

An Open Access Scholarly Journal || Index in major Databases & Metadata

CERTIFICATE OF PUBLICATION

International Journal of Scientific Research in Engineering & Management is hereby awarding this certificate to

Tanmay T. Magdum

in recognition to the publication of paper titled

Mart Billing System

published in IJSREM Journal on **Volume 08 Issue 04 April, 2024**

Editor-in-Chief
IJSREM Journal

www.ijsrem.com

ijsremjournal@gmail.com



SAMARTH KRUPA TRADERS

S.S.Teli-8975767018
S.A.Teli-9527952812

Date-

Subject:- Completion Letter for Mart Billing System.
Successfull implementation & Delivery.

Dear

I am delighted to extend my warmest congratulations on the successful completion of the Mart Billing System. It gives me immense pleasure to acknowledge the dedication, hard work, & expertise demonstrated by the entire project team, throughout the execution phase.

As the sponsor of this project, I am filled with pride & gratitude for the outstanding outcomes achieved. Your professionalism, ingenuity, & resilience have undoubtedly contributed to the project success. On behalf of our firm Samarth Krupa Trader, I extend my sincerest appreciation to each member of the project team, & Project Guide Mrs. D.A. Nikam Mam. & everyone who played a role in making this vision a reality.

Once again, congratulations on this remarkable accomplishment. May this milestone serve as a reminder of the incredible feats we can achieve when we work on real Time.

Thankyou,
With best Regards,